Motor C (v2): Forecast Dietario con LightGBM — Modelo Global con Lag Features

Aguacate Team · TFM La Dieta de un País

Estrategia

A diferencia del notebook `05_forecast_series_temporales.ipynb` (Prophet local, 210 modelos independientes), este notebook entrena **un único modelo LightGBM global** sobre todas las series combinadas.

	Prophet (nb05)	LightGBM Global (nb05b)
Modelos	210 (1 por país × macro)	1 global
Datos de entrenamiento	~13 obs por modelo	~2100 obs totales
Features	tendencia + changepoints	lags temporales + contexto país
Intervalos de confianza	nativos (Bayesianos)	quantile regression (q=0.10, q=0.90)
Forecast	directo	recursivo (cada predicción alimenta el siguiente lag)

Hipótesis: el modelo global captura patrones transversales entre países y macrocategorías que los modelos locales de Prophet no pueden ver. La ventaja es especialmente relevante para tendencias compartidas (p.ej., aumento global de azúcares, reducción de cereales en economías emergentes).

Limitación documentada (L8): forecast recursivo acumula error — las predicciones de años lejanos (2029-2030) dependen de las predicciones intermedias, no de datos reales.

In [1]:
```
import pandas as pd
import numpy as np
```

```python
import joblib
import warnings
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
warnings.filterwarnings('ignore')

import lightgbm as lgb
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from sklearn.model_selection import cross_val_score, KFold
from sklearn.preprocessing import LabelEncoder

SEED = 42
np.random.seed(SEED)

MACROS = [
    'pct_Cereales', 'pct_Tuberculos', 'pct_Azucares',
    'pct_Aceites_Grasas', 'pct_Carnes', 'pct_Lacteos_Huevos', 'pct_Frutas_Verduras'
]
FEATURES_LGB_FINAL = MACROS + ['log_Food_CPI', 'cluster_id']
HORIZON_YEAR = 2030
N_LAGS = 3  # lag1, lag2, lag3

MACRO_COLORS = {
    'pct_Cereales':       '#f0a500',
    'pct_Tuberculos':     '#8b4513',
    'pct_Azucares':       '#e84393',
    'pct_Aceites_Grasas': '#ffd700',
    'pct_Carnes':         '#c0392b',
    'pct_Lacteos_Huevos': '#a0a0a0',
    'pct_Frutas_Verduras':'#2ecc71'
}

print(f'LightGBM version: {lgb.__version__}')
print('Librerías cargadas')
```

```
LightGBM version: 4.6.0
Librerías cargadas
```

In [2]:
```
# ── CARGA DE DATOS ──────────────────────────────────────────────
df = pd.read_parquet('../processed/03_clusters.parquet')
lgb_co2_model = joblib.load('../processed/lgb_final.pkl')

# Encoder para país (feature categórica)
le_country = LabelEncoder()
df['country_enc'] = le_country.fit_transform(df['Area'])

# Encoder para macrocategoría
le_macro = LabelEncoder()

print(f'Dataset: {df.shape}')
print(f'Países: {df["Area"].nunique()} | Años: {df["Year"].min()}–{df["Year"].max()}')
print(f'Modelo CO₂ LightGBM cargado: {type(lgb_co2_model).__name__}')
```

```
Dataset: (390, 16)
Países: 30 | Años: 2010–2022
Modelo CO₂ LightGBM cargado: LGBMRegressor
```

In [3]:
```
# ── CONSTRUCCIÓN DEL DATASET DE ENTRENAMIENTO ──────────────────
# Estrategia: formato largo + lag features
# Una fila = (país, macro, año_t) con target = valor en año_t
# Features: lag1=año_(t-1), lag2=año_(t-2), lag3=año_(t-3) + contexto

def build_lag_dataset(df, macros, n_lags=3):
    records = []
    for country in df['Area'].unique():
        sub = df[df['Area'] == country].sort_values('Year').reset_index(drop=True)
        country_enc = sub['country_enc'].iloc[0]
        cluster_id  = sub['cluster_id'].iloc[-1]
        log_cpi     = np.log(sub['Food_CPI'].iloc[-1])  # último valor

        for macro in macros:
            macro_enc = le_macro.fit_transform(macros).tolist().index(
                le_macro.transform([macro])[0] if macro in le_macro.classes_ else 0
            )
            values = sub[macro].values  # 13 valores temporales
            years  = sub['Year'].values
```

```
            for i in range(n_lags, len(values)):
                row = {
                    'Area':        country,
                    'country_enc': country_enc,
                    'macro':       macro,
                    'year':        years[i],
                    'cluster_id':  cluster_id,
                    'log_Food_CPI':log_cpi,
                    'target':      values[i],
                }
                for lag in range(1, n_lags + 1):
                    row[f'lag{lag}'] = values[i - lag]
                records.append(row)

    return pd.DataFrame(records)

# Encodear macros antes de llamar a la función
le_macro.fit(MACROS)

train_df = build_lag_dataset(df, MACROS, n_lags=N_LAGS)
train_df['macro_enc'] = le_macro.transform(train_df['macro'])

print(f'Dataset entrenamiento: {train_df.shape}')
print(f'Filas por macro (media): {len(train_df) / len(MACROS):.0f}')
print(f'Distribución por macro:')
print(train_df['macro'].value_counts())
```

```
Dataset entrenamiento: (2100, 11)
Filas por macro (media): 300
Distribución por macro:
macro
pct_Cereales           300
pct_Tuberculos         300
pct_Azucares           300
pct_Aceites_Grasas     300
pct_Carnes             300
pct_Lacteos_Huevos     300
pct_Frutas_Verduras    300
Name: count, dtype: int64
```

In [4]:

```python
# ── FEATURES Y TARGET ──────────────────────────────────────────
FEAT_COLS = [f'lag{i}' for i in range(1, N_LAGS + 1)] + [
    'year', 'macro_enc', 'country_enc', 'cluster_id', 'log_Food_CPI'
]
TARGET_COL = 'target'

X_train = train_df[FEAT_COLS].values
y_train = train_df[TARGET_COL].values

print(f'X_train: {X_train.shape}')
print(f'y_train: {y_train.shape}')
print(f'Features: {FEAT_COLS}')
print(f'Target range: {y_train.min():.4f} – {y_train.max():.4f}')
```

```
X_train: (2100, 8)
y_train: (2100,)
Features: ['lag1', 'lag2', 'lag3', 'year', 'macro_enc', 'country_enc', 'cluster_id', 'log_Food_CPI']
Target range: 0.0097 – 0.7667
```

In [5]:

```python
# ── VALIDACIÓN CRUZADA – MODELO MEDIO ──────────────────────────
# Walk-forward validation: entrenamos en t=2013..2019, validamos en t=2020..2022
# (simulando que predecimos los últimos 3 años conocidos)

train_mask = train_df['year'] <= 2019
val_mask   = train_df['year'] >= 2020
```

```python
X_wf_train = train_df.loc[train_mask, FEAT_COLS].values
y_wf_train = train_df.loc[train_mask, TARGET_COL].values
X_wf_val   = train_df.loc[val_mask, FEAT_COLS].values
y_wf_val   = train_df.loc[val_mask, TARGET_COL].values

PARAMS_FORECAST = {
    'n_estimators': 400,
    'learning_rate': 0.05,
    'max_depth': 5,
    'num_leaves': 20,
    'min_child_samples': 10,
    'subsample': 0.8,
    'colsample_bytree': 0.8,
    'reg_alpha': 0.1,
    'reg_lambda': 0.1,
    'random_state': SEED,
    'verbose': -1
}

lgb_forecast_wf = lgb.LGBMRegressor(**PARAMS_FORECAST)
lgb_forecast_wf.fit(X_wf_train, y_wf_train)

y_val_pred = lgb_forecast_wf.predict(X_wf_val)

mae_wf  = mean_absolute_error(y_wf_val, y_val_pred)
r2_wf   = r2_score(y_wf_val, y_val_pred)
rmse_wf = np.sqrt(mean_squared_error(y_wf_val, y_val_pred))

print('WALK-FORWARD VALIDATION (train ≤2019, val 2020-2022):')
print(f'  R²   = {r2_wf:.4f}')
print(f'  MAE  = {mae_wf:.6f} (proporción calórica)')
print(f'  RMSE = {rmse_wf:.6f}')
print(f'  Muestras val: {len(y_wf_val)}')
```

```
WALK-FORWARD VALIDATION (train ≤2019, val 2020-2022):
  R²   = 0.9943
  MAE  = 0.006372 (proporción calórica)
  RMSE = 0.010219
  Muestras val: 630
```

In [6]:
```python
# ── MODELOS FINALES: MEDIA + CUANTILES ──────────────────────────
# Modelo medio (yhat)
lgb_forecast_mean = lgb.LGBMRegressor(**PARAMS_FORECAST)
lgb_forecast_mean.fit(X_train, y_train)

# Modelo cuantil inferior q=0.10 (IC 80% inferior)
params_q_low = {**PARAMS_FORECAST, 'objective': 'quantile', 'alpha': 0.10}
lgb_forecast_low = lgb.LGBMRegressor(**params_q_low)
lgb_forecast_low.fit(X_train, y_train)

# Modelo cuantil superior q=0.90 (IC 80% superior)
params_q_high = {**PARAMS_FORECAST, 'objective': 'quantile', 'alpha': 0.90}
lgb_forecast_high = lgb.LGBMRegressor(**params_q_high)
lgb_forecast_high.fit(X_train, y_train)

print('Modelos entrenados:')
print('  ✅ lgb_forecast_mean  (predicción central)')
print('  ✅ lgb_forecast_low   (cuantil 10% — límite inferior IC 80%)')
print('  ✅ lgb_forecast_high  (cuantil 90% — límite superior IC 80%)')
```

```
Modelos entrenados:
  ✅ lgb_forecast_mean  (predicción central)
  ✅ lgb_forecast_low   (cuantil 10% — límite inferior IC 80%)
  ✅ lgb_forecast_high  (cuantil 90% — límite superior IC 80%)
```

In [7]:
```python
# ── FORECAST RECURSIVO 2023-2030 ────────────────────────────────
# Para cada país × macro: usamos los últimos N_LAGS valores conocidos
# y predecimos año a año, alimentando cada predicción como nuevo lag.

def recursive_forecast(df_hist, country, macro, models, n_lags, horizon_year):
    """
    Genera forecast recursivo para un país × macrocategoría.
```

```
    models = (mean_model, low_model, high_model)
    """
    m_mean, m_low, m_high = models
    sub = df_hist[df_hist['Area'] == country].sort_values('Year')

    country_enc = sub['country_enc'].iloc[0]
    cluster_id  = sub['cluster_id'].iloc[-1]
    log_cpi     = np.log(sub['Food_CPI'].iloc[-1])
    macro_enc   = le_macro.transform([macro])[0]

    # Buffer de lags: valores históricos reales al inicio
    lag_buffer = list(sub[macro].values[-n_lags:])  # últimos N_LAGS valores reales

    results = []
    for year in range(2023, horizon_year + 1):
        feat = np.array([[
            lag_buffer[-1],  # lag1
            lag_buffer[-2],  # lag2
            lag_buffer[-3],  # lag3
            year,
            macro_enc,
            country_enc,
            cluster_id,
            log_cpi
        ]])

        yhat       = float(m_mean.predict(feat)[0])
        yhat_lower = float(m_low.predict(feat)[0])
        yhat_upper = float(m_high.predict(feat)[0])

        # Clamp 0-1
        yhat       = np.clip(yhat, 0, 1)
        yhat_lower = np.clip(yhat_lower, 0, 1)
        yhat_upper = np.clip(yhat_upper, 0, 1)

        results.append({
            'Area': country, 'variable': macro, 'Year': year,
            'yhat': yhat, 'yhat_lower': yhat_lower, 'yhat_upper': yhat_upper
```

```python
        })

        # Actualizar buffer con la predicción central (forecast recursivo)
        lag_buffer.append(yhat)

    return results


models_tuple = (lgb_forecast_mean, lgb_forecast_low, lgb_forecast_high)

forecast_records = []
countries = sorted(df['country_enc'].drop_duplicates().index.map(
    lambda i: df.loc[i, 'Area']
) if False else df['Area'].unique())

for i, country in enumerate(countries):
    for macro in MACROS:
        rows = recursive_forecast(df, country, macro, models_tuple, N_LAGS, HORIZON_YEAR)
        forecast_records.extend(rows)
    if (i + 1) % 10 == 0:
        print(f'  {i+1}/{len(countries)} países procesados...')

forecast_df = pd.DataFrame(forecast_records)

# Añadir años históricos (2010–2022) en formato largo para el dataset final
hist_long = df.melt(
    id_vars=['Area', 'Year'],
    value_vars=MACROS,
    var_name='variable',
    value_name='yhat'
)
hist_long['yhat_lower'] = hist_long['yhat']
hist_long['yhat_upper'] = hist_long['yhat']

full_df = pd.concat([hist_long, forecast_df], ignore_index=True)
full_df = full_df.sort_values(['Area', 'variable', 'Year']).reset_index(drop=True)

print(f'\nForecast shape (solo 2023–2030): {forecast_df.shape}')
```

```
print(f'Dataset completo (2010-2030): {full_df.shape}')
```

```
  10/30 países procesados...
  20/30 países procesados...
  30/30 países procesados...

Forecast shape (solo 2023-2030): (1680, 6)
Dataset completo (2010-2030): (4410, 6)
```

In [8]:

```
# ── CONSTRUIR VECTORES DIETARIOS Y PREDECIR CO₂ ──────────────────
pivot = forecast_df.pivot_table(
    index=['Area', 'Year'],
    columns='variable',
    values='yhat'
).reset_index()
pivot.columns.name = None

last_cluster = df.groupby('Area')['cluster_id'].last().reset_index()
pivot = pivot.merge(last_cluster, on='Area', how='left')

last_cpi = df.groupby('Area')['Food_CPI'].last().reset_index()
last_cpi['log_Food_CPI'] = np.log(last_cpi['Food_CPI'])
pivot = pivot.merge(last_cpi[['Area', 'log_Food_CPI']], on='Area', how='left')

# Re-normalizar a suma 1 (constraint composicional)
row_sums = pivot[MACROS].sum(axis=1)
pivot[MACROS] = pivot[MACROS].div(row_sums, axis=0)

pivot['CO2eq_forecast'] = lgb_co2_model.predict(pivot[FEATURES_LGB_FINAL]).astype('float32')

print(f'Pivot shape: {pivot.shape}')
print(f'CO2 proyectado - rango: {pivot["CO2eq_forecast"].min():.2f} - {pivot["CO2eq_forecast"].max():.2f} t
print(f'\nTop 5 CO2 proyectado 2030:')
print(pivot[pivot['Year'] == 2030][['Area', 'CO2eq_forecast']].sort_values('CO2eq_forecast', ascending=Fals
```

```
Pivot shape: (240, 12)
CO2 proyectado — rango: 0.81 — 9.03 t/cápita

Top 5 CO2 proyectado 2030:
          Area  CO2eq_forecast
31      Brazil        8.344073
15   Australia        6.669749
7    Argentina        6.239921
39      Canada        5.787984
167     Poland        4.394745
```

In [9]:

```python
# ── GUARDAR PARQUET ──────────────────────────────────────────
co2_fc = pivot[['Area', 'Year', 'CO2eq_forecast']]
output = full_df.merge(co2_fc, on=['Area', 'Year'], how='left')

# Guardar con sufijo _lgbm para diferenciar del Prophet
output.to_parquet('../processed/05b_forecast_lgbm.parquet', index=False)
print(f'Guardado: processed/05b_forecast_lgbm.parquet')
print(f'Shape: {output.shape}')
print(f'Columnas: {output.columns.tolist()}')
```

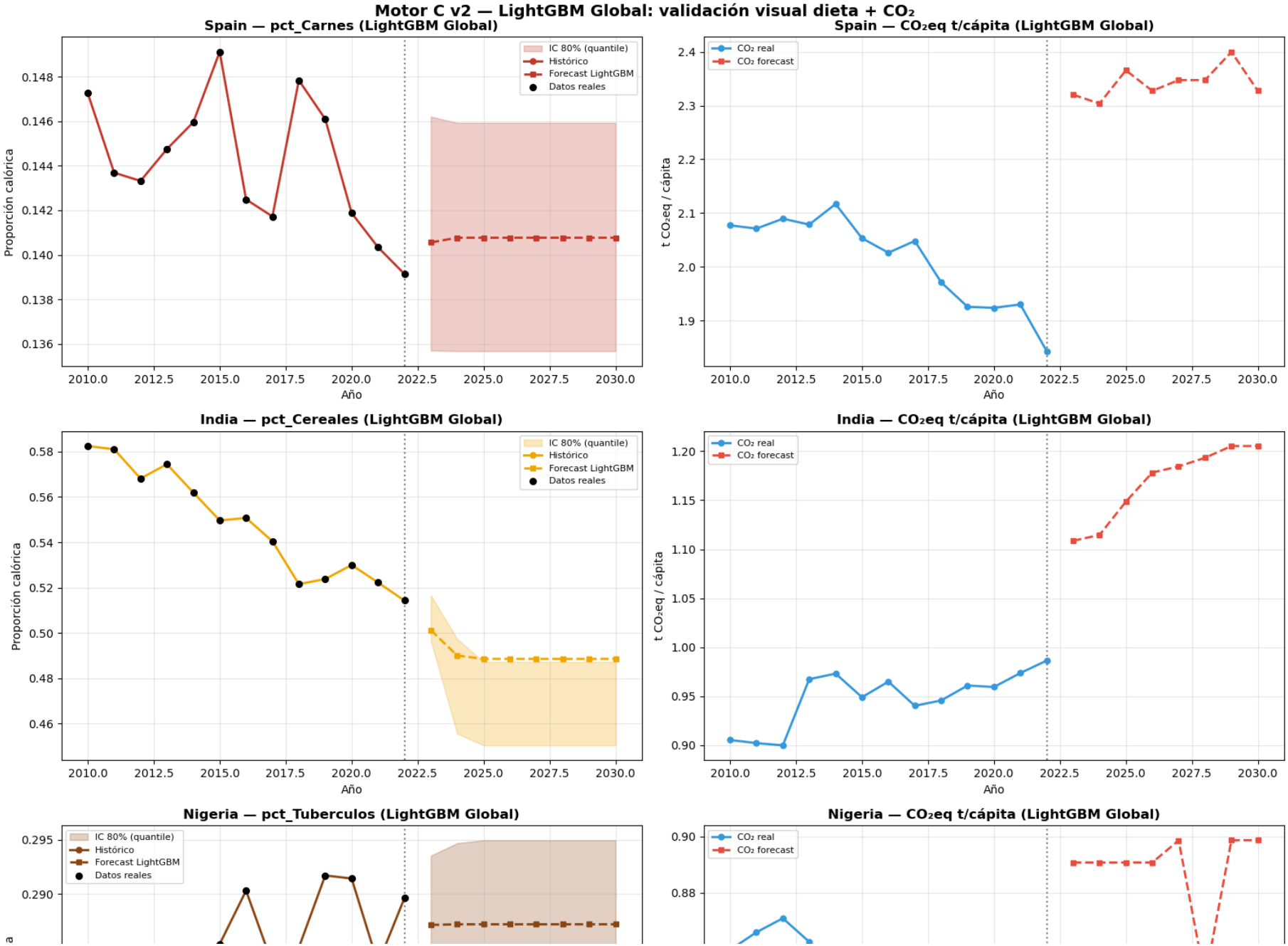
```
Guardado: processed/05b_forecast_lgbm.parquet
Shape: (4410, 7)
Columnas: ['Area', 'Year', 'variable', 'yhat', 'yhat_lower', 'yhat_upper', 'CO2eq_forecast']
```

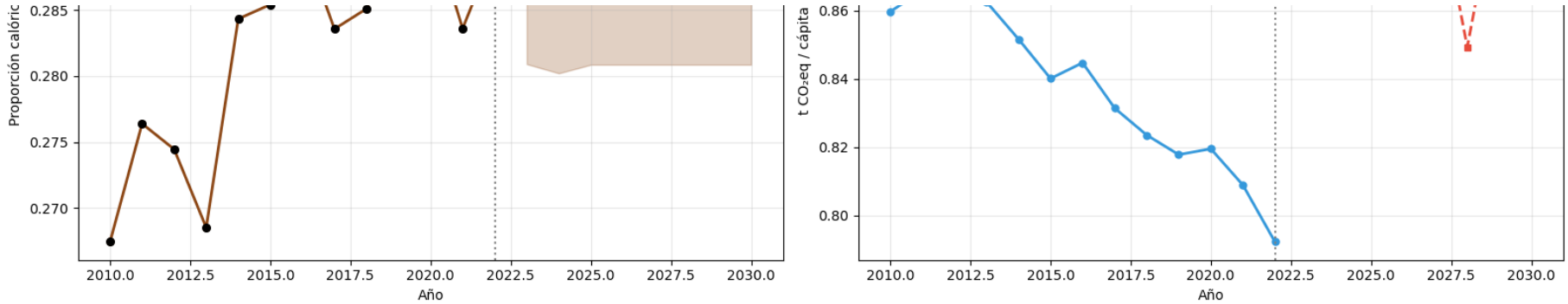
In [10]:

```python
# ── VALIDACIÓN VISUAL ────────────────────────────────────────
VALID_COUNTRIES = ['Spain', 'India', 'Nigeria']

fig, axes = plt.subplots(len(VALID_COUNTRIES), 2, figsize=(16, 5 * len(VALID_COUNTRIES)))

for row, country in enumerate(VALID_COUNTRIES):
    macro_plot = 'pct_Carnes' if country == 'Spain' else 'pct_Cereales' if country == 'India' else 'pct_Tub

    fc_country = output[(output['Area'] == country) & (output['variable'] == macro_plot)].sort_values('Year
    hist_country = df[df['Area'] == country].sort_values('Year')

    ax = axes[row, 0]
    ax.fill_between(
```

```
        fc_country[fc_country['Year'] > 2022]['Year'],
        fc_country[fc_country['Year'] > 2022]['yhat_lower'],
        fc_country[fc_country['Year'] > 2022]['yhat_upper'],
        alpha=0.25, color=MACRO_COLORS[macro_plot], label='IC 80% (quantile)'
    )
    ax.plot(fc_country[fc_country['Year'] <= 2022]['Year'],
            fc_country[fc_country['Year'] <= 2022]['yhat'],
            '-o', color=MACRO_COLORS[macro_plot], lw=2, ms=5, label='Histórico')
    ax.plot(fc_country[fc_country['Year'] > 2022]['Year'],
            fc_country[fc_country['Year'] > 2022]['yhat'],
            '--s', color=MACRO_COLORS[macro_plot], lw=2, ms=5, label='Forecast LightGBM')
    ax.scatter(hist_country['Year'], hist_country[macro_plot],
               color='black', zorder=5, s=30, label='Datos reales')
    ax.axvline(2022, color='gray', linestyle=':', lw=1.5)
    ax.set_title(f'{country} — {macro_plot} (LightGBM Global)', fontweight='bold')
    ax.set_xlabel('Año'); ax.set_ylabel('Proporción calórica')
    ax.legend(fontsize=8); ax.grid(alpha=0.3)

    # CO₂
    co2_hist = hist_country[['Year', 'CO2eq_t_per_capita']]
    co2_proj = pivot[(pivot['Area'] == country)][['Year', 'CO2eq_forecast']]

    ax2 = axes[row, 1]
    ax2.plot(co2_hist['Year'], co2_hist['CO2eq_t_per_capita'],
             '-o', color='#3498db', lw=2, ms=5, label='CO₂ real')
    ax2.plot(co2_proj[co2_proj['Year'] > 2022]['Year'],
             co2_proj[co2_proj['Year'] > 2022]['CO2eq_forecast'],
             '--s', color='#e74c3c', lw=2, ms=5, label='CO₂ forecast')
    ax2.axvline(2022, color='gray', linestyle=':', lw=1.5)
    ax2.set_title(f'{country} — CO₂eq t/cápita (LightGBM Global)', fontweight='bold')
    ax2.set_xlabel('Año'); ax2.set_ylabel('t CO₂eq / cápita')
    ax2.legend(fontsize=8); ax2.grid(alpha=0.3)

plt.suptitle('Motor C v2 — LightGBM Global: validación visual dieta + CO₂',
             fontsize=14, fontweight='bold')
plt.tight_layout()
plt.savefig('../processed/05b_forecast_lgbm_validacion.png', dpi=150, bbox_inches='tight')
plt.show()
```

```python
print('Guardado: processed/05b_forecast_lgbm_validacion.png')
```






```
Guardado: processed/05b_forecast_lgbm_validacion.png
```

In [11]:

```
# ── TABLA COMPARATIVA CO₂ 2022 vs 2030 ──────────────────────────
co2_2022 = df[df['Year'] == 2022][['Area', 'CO2eq_t_per_capita']].rename(
    columns={'CO2eq_t_per_capita': 'CO2_2022'})
co2_2030 = pivot[pivot['Year'] == 2030][['Area', 'CO2eq_forecast']].rename(
    columns={'CO2eq_forecast': 'CO2_2030_lgbm'})

resumen = co2_2022.merge(co2_2030, on='Area')
resumen['Delta_pct'] = ((resumen['CO2_2030_lgbm'] - resumen['CO2_2022']) / resumen['CO2_2022'] * 100).round
resumen = resumen.sort_values('Delta_pct')

print('CO₂ proyectado 2030 vs observado 2022 (LightGBM Global):')
print(resumen.to_string(index=False))
```

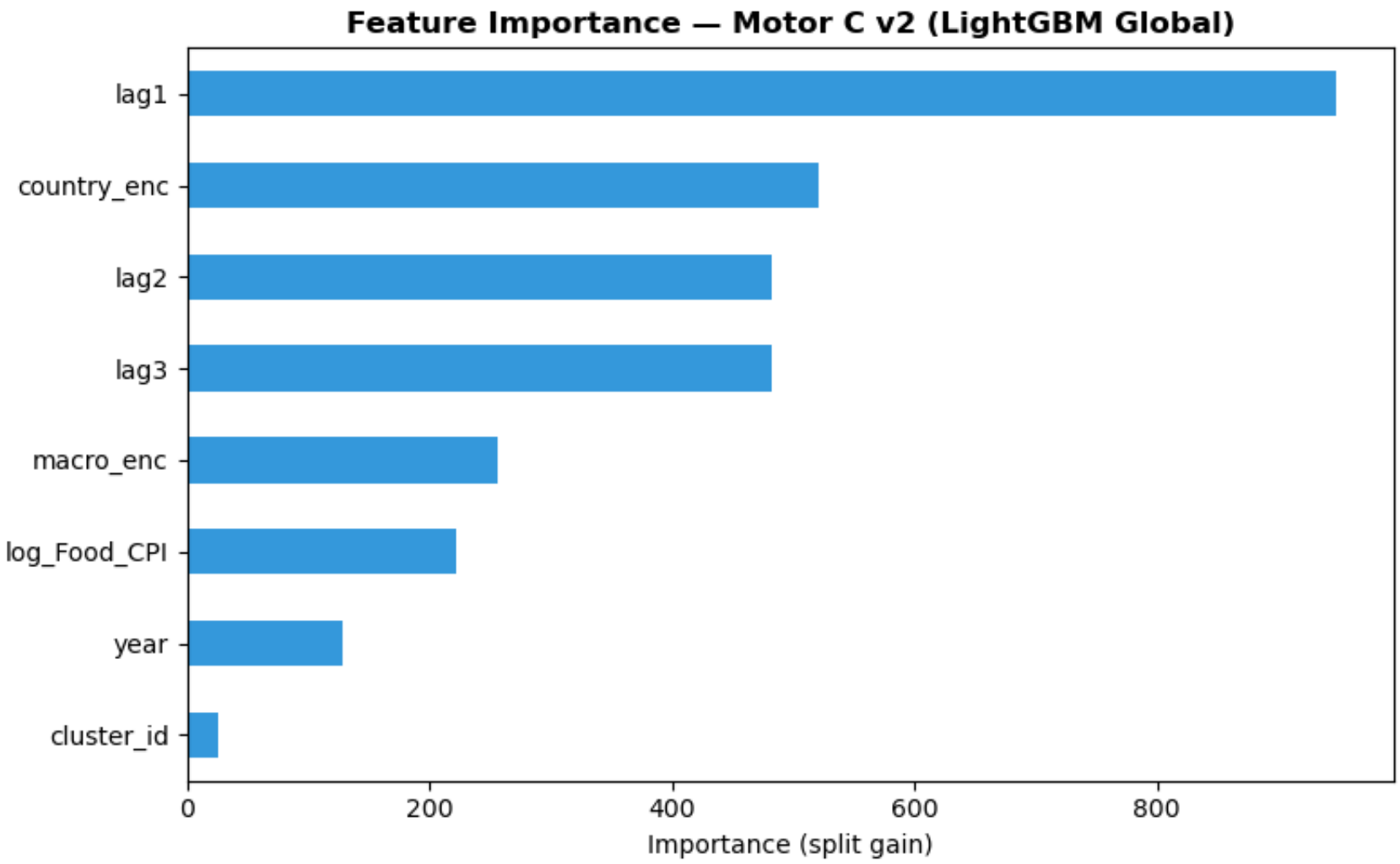
```
CO₂ proyectado 2030 vs observado 2022 (LightGBM Global):
                                                   Area  CO2_2022  CO2_2030_lgbm  Delta_pct
                                                Ethiopia  1.217610       0.881930 -27.600000
                                                  Canada  7.434406       5.787984 -22.100000
                                                Thailand  3.268874       2.613988 -20.000000
                                                  Brazil  9.748733       8.344073 -14.400000
                                               Australia  7.687373       6.669749 -13.200000
                                                 Romania  2.986566       2.614038 -12.500000
                                                 Senegal  1.184145       1.056655 -10.800000
                                            Saudi Arabia  2.354669       2.125564  -9.700000
United Kingdom of Great Britain and Northern Ireland  1.858086       1.690758  -9.000000
                                              Kazakhstan  2.779733       2.588709  -6.900000
                                                Pakistan  1.163849       1.206000   3.600000
                                                  France  2.133915       2.269848   6.400000
                                                   Kenya  1.059605       1.128004   6.500000
                                               Argentina  5.687041       6.239921   9.700000
                                                   Egypt  0.777203       0.857056  10.300000
                                              Bangladesh  0.847154       0.959488  13.300000
                                                 Nigeria  0.792288       0.898635  13.400000
                                                Colombia  3.326870       3.939178  18.400000
                                                Viet Nam  1.440029       1.719322  19.400000
                                 United States of America  3.028903       3.659689  20.799999
                                                   India  0.986313       1.205337  22.200001
                                                  Poland  3.586438       4.394745  22.500000
                                                 Germany  2.096628       2.589127  23.500000
                                                   Italy  1.746279       2.174892  24.500000
                                               Indonesia  2.042389       2.575019  26.100000
                                                   Spain  1.842866       2.327398  26.299999
                                                  Mexico  1.911802       2.425843  26.900000
                                                 Morocco  1.266582       1.674320  32.200001
                                          China, mainland  1.586632       2.818188  77.599998
                                        Republic of Korea  1.895967       4.259882 124.699997
```

In [12]:

```python
# ── IMPORTANCIA DE FEATURES ──────────────────────────────────────
feat_imp = pd.Series(
    lgb_forecast_mean.feature_importances_,
    index=FEAT_COLS
).sort_values(ascending=False)
```

```python
print('Feature importances (LightGBM Global Forecast):')
print(feat_imp.round(1))

fig, ax = plt.subplots(figsize=(8, 5))
feat_imp.plot(kind='barh', ax=ax, color='#3498db')
ax.invert_yaxis()
ax.set_title('Feature Importance — Motor C v2 (LightGBM Global)', fontweight='bold')
ax.set_xlabel('Importance (split gain)')
plt.tight_layout()
plt.savefig('../processed/05b_feature_importance.png', dpi=150, bbox_inches='tight')
plt.show()
print('Guardado: processed/05b_feature_importance.png')
```

```
Feature importances (LightGBM Global Forecast):
lag1            948
country_enc     521
lag2            483
lag3            482
macro_enc       256
log_Food_CPI    222
year            128
cluster_id       26
dtype: int32
```




```
Guardado: processed/05b_feature_importance.png
```

In [13]:

```python
# ── RESUMEN EJECUTIVO ──────────────────────────────────────────
print('=' * 60)
print('RESUMEN EJECUTIVO — MOTOR C v2 (LightGBM Global)')
print('Aguacate Team · TFM La Dieta de un País')
print('=' * 60)
print()
print('CONFIGURACIÓN:')
```

```python
print(f'  Modelos entrenados: 3 (mean + q10 + q90)')
print(f'  Features: {FEAT_COLS}')
print(f'  Lags: {N_LAGS} | Muestras entrenamiento: {X_train.shape[0]}')
print()
print('MÉTRICAS WALK-FORWARD (val 2020-2022):')
print(f'  R²   = {r2_wf:.4f}')
print(f'  MAE  = {mae_wf:.6f} (proporción calórica)')
print(f'  RMSE = {rmse_wf:.6f}')
print()
print('OUTPUTS:')
print('  processed/05b_forecast_lgbm.parquet')
print('  processed/05b_forecast_lgbm_validacion.png')
print('  processed/05b_feature_importance.png')
```

```
============================================================
RESUMEN EJECUTIVO — MOTOR C v2 (LightGBM Global)
Aguacate Team · TFM La Dieta de un País
============================================================

CONFIGURACIÓN:
  Modelos entrenados: 3 (mean + q10 + q90)
  Features: ['lag1', 'lag2', 'lag3', 'year', 'macro_enc', 'country_enc', 'cluster_id', 'log_Food_CPI']
  Lags: 3 | Muestras entrenamiento: 2100

MÉTRICAS WALK-FORWARD (val 2020-2022):
  R²   = 0.9943
  MAE  = 0.006372 (proporción calórica)
  RMSE = 0.010219

OUTPUTS:
  processed/05b_forecast_lgbm.parquet
  processed/05b_forecast_lgbm_validacion.png
  processed/05b_feature_importance.png
```

Próximo paso: comparación con Prophet (nb05)

Comparar visualmente los forecasts de ambos notebooks para los mismos países y macrocategorías. Criterios de decisión:

Criterio	Favorable a Prophet	Favorable a LightGBM Global
MAE walk-forward	si LGB $R^2 < 0.6$	si LGB $R^2 > 0.7$
Coherencia tendencial	si Prophet sigue mejor la inercia visual	si LGB captura mejor los quiebres
Amplitud IC	si LGB IC es más estrecho y creíble	si Prophet IC son demasiado amplios
Interpretabilidad	Prophet: narrativa tendencia	LGB: feature importance lag1 dominante = inercia

Decisión final documentada en memoria v1.5